Voltage: energy per charge

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Learning target: Interpret potential difference as energy transferred per unit charge.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 1.*
Repulsion: each force points away from the other object.

02. Worked example: explain the model
Higher potential by 3 V means potential-energy increase 6 J for +2 C.

03. Read and interpret the model
6 V supplies twice as much energy per coulomb.

04. Apply the physics
12 J.

05. Explain your reasoning
9 V means 9 J/C; total charge capacity and total stored energy require additional information.

06. Review: connect an earlier idea *Review: Unit 6, Day 10.*
Correct speed=2×60/0.6=200 m/s. The student used half the traveled path and gets half the speed.

07. Exit check
4 J per coulomb.

08. STRETCH: test the model
6 and 18 J; equal voltage does not imply equal total energy.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 1.*
Polarization redistributes charge. Nearer opposite charge can attract more strongly than farther like charge repels; net charge may remain zero.

02. Worked example: explain the model
Higher potential by 3 V means potential-energy increase 6 J for +2 C.

03. Read and interpret the model
6 V supplies twice as much energy per coulomb.

04. Apply the physics
12 V; 15 J.

05. Explain your reasoning
9 V means 9 J/C; total charge capacity and total stored energy require additional information.

06. Review: connect an earlier idea *Review: Unit 6, Day 10.*
True speed 340 m/s; incorrect speed≈272 m/s, biased low.

07. Exit check
6 J and 12 J; 18 J total.

08. STRETCH: test the model
ΔU=-100 microjoules; field work +100 microjoules. Negative charge loses potential energy moving to higher potential.